

12.421 / 12.621 Lecture 5

Reflection and transmission of EM waves
at oblique incidence

Topics covered

- Reflection and transmission at oblique incidence

EM BCs for zero charge and zero current at interface

$$1. \epsilon_1' \underline{E}_1^\perp = \epsilon_2' \underline{E}_2^\perp \quad \text{--or--} \quad \hat{a} \cdot (\epsilon_1' \underline{E}_1 - \epsilon_2' \underline{E}_2) = 0$$

normal component of $\epsilon' \underline{E}$ is continuous

$$2. \mu_1 \underline{H}_1^\perp = \mu_2 \underline{H}_2^\perp \quad \text{--or--} \quad \hat{a} \cdot (\mu_1 \underline{H}_1 - \mu_2 \underline{H}_2) = 0$$

normal component of $\mu \underline{H}$ is continuous

$$3. \underline{E}_1^\parallel = \underline{E}_2^\parallel \quad \text{--or--} \quad \hat{a} \times (\underline{E}_1 - \underline{E}_2) = \underline{0}$$

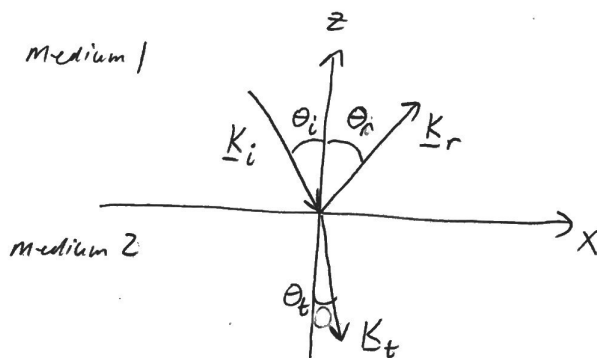
tangential component of $\underline{E}$ is continuous

$$4. \underline{H}_1^\parallel = \underline{H}_2^\parallel \quad \text{--or--} \quad \hat{a} \times (\underline{H}_1 - \underline{H}_2) = \underline{0}$$

tangential component of $\underline{H}$ is continuous

Reflection and transmission at oblique incidence

• Let's start with the basics: "geometric optics"



where $\underline{k}_i \equiv$ wave number
vector of incident
wave (and similarly
for $\underline{k}_r$ and $\underline{k}_t$)

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- Fields for a uniform plane wave. (position vector $\underline{r}$)

$$\left. \begin{aligned} \underline{E}^i(\underline{r}, t) &= \underline{E}_0^i e^{j(\underline{k}_i \cdot \underline{r} - \omega t)} \\ \underline{H}^i(\underline{r}, t) &= \frac{1}{\eta_1} (\hat{\underline{k}}_i \times \underline{E}^i) \end{aligned} \right\} \text{incident}$$

$$\left. \begin{aligned} \underline{E}^r(\underline{r}, t) &= \underline{E}_0^r e^{j(\underline{k}_r \cdot \underline{r} - \omega t)} \\ \underline{H}^r(\underline{r}, t) &= \frac{1}{\eta_1} (\hat{\underline{k}}_r \times \underline{E}^r) \end{aligned} \right\} \text{reflected}$$

$$\left. \begin{aligned} \underline{E}^t(\underline{r}, t) &= \underline{E}_0^t e^{j(\underline{k}_t \cdot \underline{r} - \omega t)} \\ \underline{H}^t(\underline{r}, t) &= \frac{1}{\eta_2} (\hat{\underline{k}}_t \times \underline{E}^t) \end{aligned} \right\} \text{transmitted}$$

- Recall that ω is the same everywhere and for all time and set by the source

$\Rightarrow$ dispersion relation gives

$$\omega = k_i v_{p1} = k_r v_{p1} = k_t v_{p2}$$

$$\Rightarrow k_i = k_r = \frac{v_{p2}}{v_{p1}} k_t = \frac{v_{p2}}{v_{p1}} \frac{c \sqrt{\epsilon_1}}{c \sqrt{\epsilon_2}} k_t = \frac{n'_1}{n'_2} k_t$$

- Total $\underline{E}$ and $\underline{H}$ fields in medium 1 related to those in medium 2 through BCs
Has generic structure

$$B_i^i e^{j(\underline{k}_i \cdot \underline{r} - \omega t)} + B_r^r e^{j(\underline{k}_r \cdot \underline{r} - \omega t)} = B_t^t e^{j(\underline{k}_t \cdot \underline{r} - \omega t)}$$

Don't worry about B coeffs for now, just note that the only space and time dependence is in exponents

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- BCs must hold everywhere on xy plane at the interface $z=0$ for all times

$\Rightarrow$ exponential factors must be equal at $z=0$

- time factors already equal because $\omega = \text{const}$
- spatial components give

$$\underline{K}_i \cdot \underline{r} = \underline{K}_r \cdot \underline{r} = \underline{K}_t \cdot \underline{r} \quad @ \quad z=0$$

expanding:

$$x K_{ix} + y K_{iy} = x K_{rx} + y K_{ry} = x K_{tx} + y K_{ty}$$

for all x and y $\Rightarrow$ only holds if

components are separately equal

$$\hookrightarrow \text{for } x=0: \quad K_{iy} = K_{ry} = K_{ty}$$

$$\hookrightarrow \text{for } y=0: \quad K_{ix} = K_{rx} = K_{tx}$$

- We are free to choose our coordinate system; for convenience, let's define $\underline{K}_i$ to lie in the x - z plane

$$\Rightarrow K_{iy} = K_{ry} = K_{ty} = 0$$

and the only non-zero components of the wavenumber vector lie in the x - z plane for the incident, reflected, and transmitted waves

- From the above, we get

1st law of geometric optics:

The incident, reflected, and transmitted wave vectors (rays) form a plane called the plane of incidence, which also includes the normal to the surface (in our case $\hat{z}$)

- Now, given that $K_{ix} = K_{rx} = K_{tx}$

$$K_i \sin \theta_i = K_r \sin \theta_r = K_t \sin \theta_t$$

$\theta_i \equiv$ angle of incidence

$\theta_r \equiv$ angle of reflection

$\theta_t \equiv$ angle of refraction (or transmission)

} all measured
w.r.t. normal

and from dispersion relation we have $K_i = K_r$

$$\boxed{\theta_i = \theta_r \equiv \theta_i}$$

2nd law of geometric optics, commonly called law of reflection: angle of incidence $\theta_i =$ angle of reflection θ_r

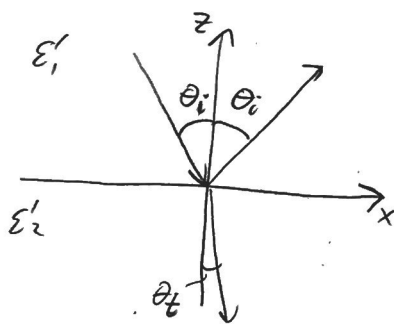
and from dispersion relation we have

$$K_i = K_r = K_t = \frac{v_{p2}}{v_{p1}} K_t = \frac{Re(\sqrt{\epsilon_1})}{Re(\sqrt{\epsilon_2})} K_t = \frac{n_1'}{n_2'} K_t$$

3rd law of g.o., a.k.a. law of refraction or Snell's Law

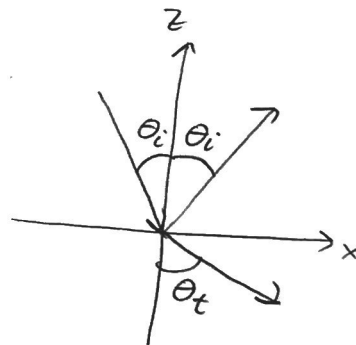
$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{v_{p2}}{v_{p1}} = \frac{n_1'}{n_2'} = \frac{Re(\sqrt{\epsilon_1})}{Re(\sqrt{\epsilon_2})}$$

- Note that up to this point, we did not invoke any EM principles to derive the 3 laws of geometric optics $\Rightarrow$ These laws are valid for all wave types (water, sound, seismic, etc.) and many of you will have seen them before
 - Before moving on to EM waves, let's quickly consider implications of Snell's Law
- Let's keep it simple and consider lossless media



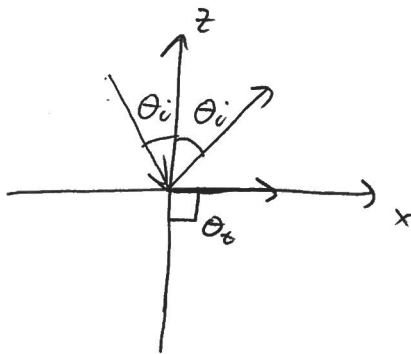
$$\epsilon'_1 < \epsilon'_2 \Rightarrow \theta_t < \theta_i$$

"Inward refraction"



$$\epsilon'_1 > \epsilon'_2 \Rightarrow \theta_t > \theta_i$$

"Outward refraction"



critical angle $\theta_i = \theta_* \rightarrow \theta_t = \frac{\pi}{2}$

to find θ_* , apply Snell's Law

$$\sin \theta_* = \frac{n_2}{n_1} \sin \theta_t \Big|_{\theta_t = \frac{\pi}{2}} = \frac{n_2}{n_1}$$

$$\Rightarrow \theta_* = \sin^{-1} \left(\frac{n_2(\sqrt{\epsilon'_2})}{n_1(\sqrt{\epsilon'_1})} \right)$$

when $\theta_i = \theta_*$ refracted wave travels along the boundary

when $\theta_i > \theta_*$ (total reflection) (only possible when $n'_1 > n'_2$)
 $\rightarrow$ gives diamonds sparkle

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• Reflection and transmission at oblique angles (cont.)

- Now that we have an okay grasp of the exponential terms, let's return to EM waves

- Applying the BCs discussed before

$$1. \epsilon'_1 (E_{0z}^i + E_{0z}^r) = \epsilon'_2 E_{0z}^t$$

from:

Gauss's Law (E)

$$2. \mu_1 (H_{0z}^i + H_{0z}^r) = \mu_2 H_{0z}^t$$

Gauss's Law for mag

$$3. E_{0x}^i + E_{0x}^r = E_{0x}^t$$

$$E_{0y}^i + E_{0y}^r = E_{0y}^t$$

Farraday's Law

$$4. H_{0x}^i + H_{0x}^r = H_{0x}^t$$

$$H_{0y}^i + H_{0y}^r = H_{0y}^t$$

Ampere's Law

- Now, we need to get organized by defining polarizations

• What's a good reference for polarization?

plane of incidence (see 1st law of geo. optics)

⇒ two commonly defined polarizations

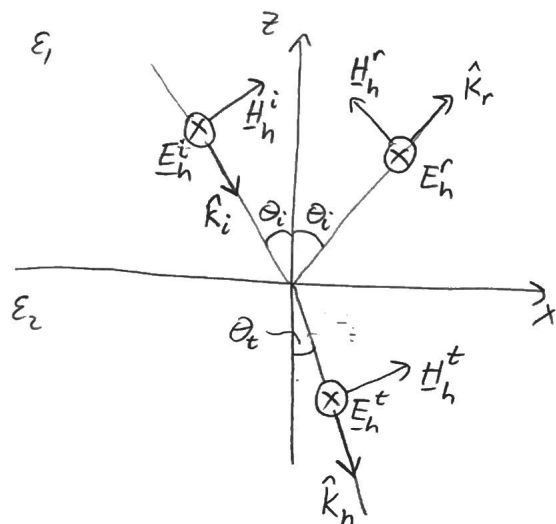
1. perpendicular or horizontal → E field is normal to plane of incidence

2. parallel or vertical → H field is normal to plane of incidence

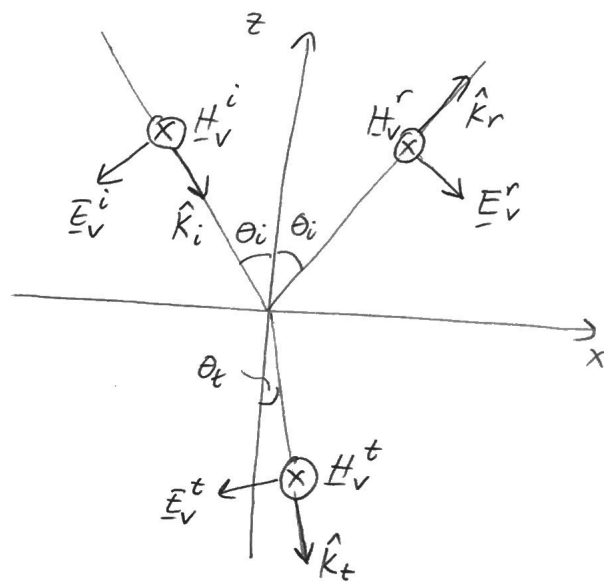
(so named because E field lies in plane of incidence and usually has a vertical component)

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- Polarization definitions (cont.)



Perpendicular or horizontal (H)



parallel or vertical (V)

- As we discussed in the lecture on polarization, any polarization can be decomposed into orthogonal linear polarizations

=> Hereafter, only need to consider horizontal (H) and vertical (V) (same as perpendicular and parallel, respectively)

- H polarization

$$\begin{aligned}
 \underline{E}_h^i &= \hat{y} E_{h0}^i e^{-\gamma_1(x \sin \theta_i - z \cos \theta_i)} \\
 \underline{H}_h^i &= (\hat{x} \cos \theta_i + \hat{z} \sin \theta_i) \frac{E_{h0}^i}{\eta_1} e^{-\gamma_1(x \sin \theta_i - z \cos \theta_i)} \\
 \underline{E}_h^r &= \hat{y} E_{h0}^r e^{-\gamma_1(x \sin \theta_i + z \cos \theta_i)} \\
 \underline{H}_h^r &= (-\hat{x} \cos \theta_i + \hat{z} \sin \theta_i) \frac{E_{h0}^r}{\eta_1} e^{-\gamma_1(x \sin \theta_i + z \cos \theta_i)} \\
 \underline{E}_h^t &= \hat{y} E_{h0}^t e^{-\gamma_2(x \sin \theta_t - z \cos \theta_t)} \\
 \underline{H}_h^t &= (\hat{x} \cos \theta_t + \hat{z} \sin \theta_t) \frac{E_{h0}^t}{\eta_2} e^{-\gamma_2(x \sin \theta_t - z \cos \theta_t)}
 \end{aligned}
 \left. \begin{array}{l} \\ \\ \\ \\ \end{array} \right\} \begin{array}{l} \text{incident} \\ \\ \text{reflected} \\ \\ \text{transmitted} \end{array}$$

- H pol (cont)

apply BCs

• From Gauss's Law

$$\epsilon_1' (E_{0z}^i + E_{0z}^r) = \epsilon_2' E_{0z}^t$$

But $E_{0z}^i = E_{0z}^r = E_{0z}^t = 0$, so no information

• From Gauss's Law for mag

$$H_{0z}^i + H_{0z}^r = H_{0z}^t$$

$$\frac{E_{h0}^i}{\mu_1} \sin \theta_i + \frac{E_{h0}^r}{\mu_1} \sin \theta_i = \frac{E_{h0}^t}{\mu_2} \sin \theta_t$$

Apply Snell's and cancel terms, we get

$$\underbrace{E_{h0}^i + E_{h0}^r}_{\text{you don't see how we get here}} = E_{h0}^t \quad (\text{work this out on your own if})$$

→ tangential component of E field is conserved;
same condition as we get from Faraday's Law

• From Ampere's Law

$$H_{0x}^i + H_{0x}^r = H_{0x}^t \quad (\text{y components are all zero})$$

$$\Rightarrow \frac{E_{h0}^i}{\mu_1} \cos \theta_i - \frac{E_{h0}^r}{\mu_1} \cos \theta_i = \frac{E_{h0}^t}{\mu_2} \cos \theta_t$$

$$\Rightarrow E_{h0}^t = \frac{\mu_2 \cos \theta_i}{\mu_1 \cos \theta_t} (E_{h0}^i - E_{h0}^r)$$

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- H pol. (cont)

- As we did last lecture, let us define the reflection and transmission coeffs.

$$\rho_h = \frac{E_{h0}^r}{E_{h0}^i} \quad \text{and} \quad \tau_h = \frac{E_{h0}^t}{E_{h0}^i}$$

- From our BCs, we have

$$1 + \rho_h = \tau_h \quad \text{and} \quad \frac{\eta_2 \cos \theta_i}{\eta_1 \cos \theta_t} (1 - \rho_h) = \tau_h$$

$$\Rightarrow \rho_h = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} = \frac{\cos \theta_i - \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}$$

↑
apply Snell's Law

and

$$\tau_h = \frac{2 \eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} = \frac{2 \cos \theta_i}{\cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}$$

where ρ_h and τ_h are the Fresnel reflection and transmission coefficients, respectively

- We can see straight away that ρ_h and τ_h reduce to ρ and τ for normal incidence when $\theta_i = \theta_t = 0$

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- V polarization

- To get the V polarized Fresnel coefficients R and T_V , simply swap $\underline{E}$ with $\underline{H}$ and $\underline{H}$ with $-\underline{E}$ and plug and chug
- Do this exercise on your own
- Results

Fresnel coefficients for V polarization

$$R_V = \frac{E_{V_o}^r}{E_{V_o}^i} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

$$= \frac{\sqrt{\left(\frac{\epsilon_2}{\epsilon_1}\right) - \sin^2 \theta_i} - \frac{\epsilon_2}{\epsilon_1} \cos \theta_i}{\sqrt{\left(\frac{\epsilon_2}{\epsilon_1}\right) - \sin^2 \theta_i} + \frac{\epsilon_2}{\epsilon_1} \cos \theta_i}$$

$$T_V = \frac{E_{V_o}^t}{E_{V_o}^i} = (1 + R) \frac{\cos \theta_i}{\cos \theta_t} = \frac{2 \eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

- Reflected and transmitted power

- Often interested in the power reflected and transmitted

↳ Poynting vector gives energy flux (power per unit area)

↳ From discussion last lecture,

$$S_h^K = \frac{|E_{h_o}^K|^2}{2 \eta_1} \cos \phi_{\eta_1} \quad \text{for } K=i, r \quad \text{and} \quad S_h^t = \frac{|E_{h_o}^t|^2}{2 \eta_2} \cos \phi_{\eta_2}$$

↑
phase at η_1 (note change of variable from previous lectures)

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- Power (cont)

↳ because $\theta_i \neq \theta_t$, we need to consider average power over an area A illuminated by incident beam on the interface

$$A_i = A_r = A \cos \theta_i \quad \text{and} \quad A_t = A \cos \theta_t$$

$$\Rightarrow P_h^i = S_h^i A_i \quad P_h^r = S_h^r A_r \quad P_h^t = S_h^t A_t$$

• Define reflectivity Γ (also called reflectance) as ratio of reflected power to incident power

$$\boxed{\Gamma_h = \frac{P_h^r}{P_h^i} = \frac{|E_{h0}^r|^2}{|E_{h0}^i|^2} = |\rho_h|^2}$$

Similarly

$$\boxed{\Gamma_v = |\rho_v|^2}$$

• Define transmissivity T (also called transmittance) as ratio of transmitted power to incident power

$$T_h = \frac{P_h^t}{P_h^i} = \frac{|E_{h0}^t|^2}{|E_{h0}^i|^2} \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} = |\tau_h|^2 \Phi$$

$$T_v = |\tau_v|^2 \Phi$$

$$\text{where } \Phi = \frac{|\eta_1| \cos \theta_t \cos \phi \eta_2}{|\eta_2| \cos \theta_i \cos \phi \eta_1}$$

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Special considerations for oblique incidence

1. In lossy media ($\sigma > 0$), there will be a complex-valued angle of refraction, call it $\tilde{\theta}_t$

* only the real component θ_t has physical meaning.

but to see how $\tilde{\theta}_t$ arises, recall that the exponential terms must be equal at the interface ($z=0$),

requiring $\gamma_2 \sin \tilde{\theta}_t = \gamma_1 \sin \theta_i$

$$\Rightarrow \sin \tilde{\theta}_t = \frac{\alpha_1 + j\beta_1}{\alpha_2 + j\beta_2} \sin \theta_i$$

$\sin \theta_i$ is always real-valued, so if either medium is conductive, the respective attenuating constant (α) is nonzero and $\tilde{\theta}_t$ is complex valued (β always nonzero)

2. Perfect conductor, perfect reflector

• recall that

$$P_h = \frac{\sqrt{E_1} \cos \theta_i - \sqrt{E_2} \cos \theta_t}{\sqrt{E_1} \cos \theta_i + \sqrt{E_2} \cos \theta_t}$$

$$\text{and } \tilde{\epsilon}_h = 1 + P_h$$

• A perfect conductor $\epsilon_2'' \rightarrow \infty$

$\Rightarrow P_h \approx -1$ and $\tilde{\epsilon}_h \approx 0$ (similarly for V polarization)

$\Rightarrow$ total, out-of-phase, reflection

• Useful for calibration, which often uses corner reflectors made of good conductors

Special considerations for oblique incidence (cont.)

3. Brewster angle

- recall

$$R_V = \frac{\sqrt{\epsilon_1} \cos \theta_t - \sqrt{\epsilon_2} \cos \theta_i}{\sqrt{\epsilon_1} \cos \theta_t + \sqrt{\epsilon_2} \cos \theta_i} \quad \text{and} \quad T_V = (1 + R_V) \frac{\cos \theta_i}{\cos \theta_t}$$

- When

$$\sqrt{\epsilon_1} \cos \theta_t = \sqrt{\epsilon_2} \cos \theta_i, \quad R_V = 0 \Rightarrow \text{total transmission into medium 2 for V-polarized waves}$$

- Note that equality can only hold if both materials are non-conductive

- So, define Brewster angle θ_B as incidence angle that allows for total transmission of V polarized EM waves

$$\theta_B = \tan^{-1} \left(\sqrt{\frac{\epsilon_2'}{\epsilon_1'}} \right)$$

- Now, recall H polarized reflection coef

$$R_H = \frac{\sqrt{\epsilon_1} \cos \theta_i - \sqrt{\epsilon_2} \cos \theta_t}{\sqrt{\epsilon_1} \cos \theta_i + \sqrt{\epsilon_2} \cos \theta_t}$$

Why is there no equivalent to the Brewster angle for H polarization?